Chapter 10

Education, Health and Everyday Life

In June and July 2024, first-year senior secondary students from nine public schools in Benin City, Nigeria, took part in an after-school programme to practise English. They worked in pairs with Microsoft Copilot, a chatbot built on GPT-4. Teachers set the topic and the opening prompt. The students then questioned the chatbot freely for 90 minutes, up to twice a week, for six weeks. Places were allocated by lottery among the students who volunteered, which turned the programme into a randomised controlled trial. In a working paper, the World Bank team that evaluated it reported a gain of 0.31 standard deviations on an end-of-programme assessment, and 0.23 standard deviations on the English component alone (De Simone et al., 2025).

A few months earlier, in autumn 2023, a field experiment in a large high school in Turkey gave nearly 1,000 mathematics students access to GPT-4 during practice sessions. Students with an unrestricted chatbot solved more practice problems, then scored 17% lower than students without access when they sat an exam on their own (Bastani et al., 2025). The two studies used similar technology and reached opposite headline results. The difference lies in how the tools were set up, what was measured and when, and the same questions decide how to read the evidence on AI in clinics and in private life. Effects on jobs and productivity are treated in Chapter 8, the general taxonomy of AI risks in Chapter 11, and the enforcement of data-protection law against AI developers in Chapter 13.

10.1 AI tutors: the benchmark and the trials

The hope behind AI tutoring has a long pedigree. In 1984 the educational psychologist Benjamin Bloom reported that students taught one-to-one by a tutor performed about two standard deviations better than students taught in a conventional class. He called the search for group methods that could match this the “2 sigma problem” (Bloom, 1984). Later research confirmed that tutoring works, but

with much smaller effects. Kurt VanLehn's 2011 review of experiments comparing human tutoring, computer tutoring and no tutoring found an effect of 0.79 standard deviations for human tutors, far below Bloom's figure. Intelligent tutoring systems, the rule-based programs that give feedback on each step of a solution, reached 0.76, nearly as much (VanLehn, 2011). A meta-analysis of 50 controlled evaluations of such systems found a median effect of 0.66 standard deviations, but the effect depended heavily on whether learning was measured on tests developed locally for the evaluation or on standardised ones (Kulik & Fletcher, 2016).

The best current estimate for human tutoring in schools comes from a meta-analysis of randomised field experiments in pre-school to year 12. It found a pooled effect of 0.288 standard deviations. Effects were largest for programmes that used teachers or paraprofessionals as tutors, took place in earlier grades, met at least three days a week and ran during the school day (Nickow et al., 2024). The obstacle has always been cost: a trained tutor for every pupil several times a week is unaffordable for most school systems. Large language models (LLMs, the text-generating systems described in Chapter 3) revived the question of whether software could supply part of what a tutor provides at a fraction of the price.

Going deeper: How to read an effect size and an education trial

Education studies report results in standard deviations (SD) of the test score so that different tests can be compared. If scores are roughly normally distributed, an effect of 0.3 SD moves the average treated student from the 50th to about the 62nd percentile of the comparison group, and 0.63 SD to about the 74th. The psychologist Jacob Cohen's conventions, which call 0.2 "small" and 0.8 "large", come from laboratory psychology and mislead in schools.

In a 2019 working paper, later published in *Educational Researcher*, Matthew Kraft assembled effect sizes from 747 randomised trials of education interventions with standardised achievement outcomes. Their median was 0.10 SD, and he proposed treating less than 0.05 as small, 0.05 to just under 0.20 as medium and 0.20 or more as large (Kraft, 2019; published as Kraft, 2020). By that yardstick, all the AI tutoring results below are large, which is a reason for both interest and suspicion.

Suspicion, because study design inflates effect sizes. In 645 studies drawn from 12 reviews, Cheung and Slavin found effects roughly twice as large in published articles as in unpublished reports, in small trials as in large ones, and on measures made by the experimenters as on independent tests. Quasi-experiments, which compare groups not formed by chance, produced larger effects than randomised experiments (Cheung & Slavin, 2016). How much weight a trial deserves also depends on how it assigned participants, what it estimated and who dropped out. Assigning whole classes to arms, as in the Turkish trial, prevents students from sharing tools across arms; but students in the same class resemble one another, so the effective sample is smaller than the head count. The estimand is the precise quantity a study sets out to estimate. An "intention-to-treat" estimate compares everyone offered a programme with everyone who was not, whether or not they attended. It answers the policy question "what happens if we offer this?" Attrition matters because if weaker students in one arm skip the final test, the remaining groups are no longer comparable.

Table 10.1 summarises the main randomised studies of LLM-based tutoring published by October 2026. The Nigerian programme is one of the most discussed because it took place in a lower-income country. About half of the 1,328 volunteers were offered places; the rest formed a control group that continued normal lessons (De Simone et al., 2025). The authors estimate that the six-week programme produced gains equivalent to 1.5 to 2 years of ordinary schooling, at about US$48 per pupil (US$9 at the margin), which would place it among the

more cost-effective education interventions evaluated in developing countries. Effects were positive across the range of prior attainment but largest for girls and for students who were already doing well; the authors caution that the result for girls may partly reflect one girls-only school. Attendance averaged 72%, so effects for students who actually attended are larger than the intention-to-treat figures. More control than treated students missed the final test (337 of the controls sat it, against 422 of those offered places), and the authors correct for this statistically. The paper has not been peer-reviewed.

Table 10.1: Controlled studies of LLM-based tutoring with random or matched assignment (see the design column). Effect sizes are measured on different tests and are not directly comparable. Source: (Bastani et al., 2025; De Simone et al., 2025; Henkel et al., 2024; Kestin et al., 2025; J. Liu et al., 2026; Wang et al., 2024).

Study	Setting and sample	Design	AI set-up	Result
De Simone et al. (working paper)	Nigeria, 9 public schools, 2024; 1,328 volunteers	Individual lottery; 6 weeks after school	Copilot (GPT-4), teacher-designed prompts, pairs	+0.31 SD overall, +0.23 SD English (intention-to-treat)
Kestin et al.	Harvard introductory physics, 2023; 194 students	Crossover: each student had both conditions	Tutor fed instructors' step-by-step solutions	+0.63 SD versus in-class active learning
Henkel et al. (preprint)	Ghana, 11 schools, grades 3–9; about 1,000 students	Schools assigned to arms; 8 months, two 30-minute sessions a week	Rori, maths tutor on WhatsApp	+0.37 SD on maths growth scores
Wang et al. (preprint)	US online tutoring; 900 tutors, 1,800 students	Tutors randomised	AI suggests moves to human tutors	+4 percentage points topic mastery; +9 for lower-rated tutors
Bastani et al.	Turkey, one high school, 2023; nearly 1,000 students in about 50 classes	Classes assigned to three arms; students randomly assigned to classes	GPT-4 unrestricted vs hint-only tutor	Exam −17% (unrestricted); no significant change (hint-only)

Study	Setting and sample	Design	AI set-up	Result
Liu et al. (working paper)	US public university, autumn 2025; 2,379 undergraduates, 30 instructors	Instructors randomised within blocks of similar courses; one semester	GPT-4o tutor drawing on instructor-selected course materials	Final grades −0.37 SD within same-course blocks; not significant in full sample

The Harvard trial tested a different question: could a carefully built AI tutor beat a good lesson, not merely no lesson? Kestin and colleagues compared an AI tutor, used at home, with an in-class "active learning" session in an introductory physics course for life-science students in autumn 2023 (Kestin et al., 2025). In the crossover design, 194 students each took one lesson in each format, on two topics, so that every student served as their own comparison. A regression adjusting for prior knowledge gave an effect of 0.63 SD, which the authors consider an underestimate because many students reached the top of the scale. Students using the tutor spent a median of 49 minutes, against 60 minutes in class, and rated their engagement higher. The tutor was no general chatbot: it was given step-by-step solutions written by the instructors and a system prompt built on research about how students learn. The material was introductory and new to the students, and the lessons targeted understanding and application. Some of the authors developed the active-learning methods used as the comparison, which makes it a demanding benchmark but also gives the team a stake in both arms.

In Ghana, Rori, a maths tutor that runs on WhatsApp on basic phones, was used by about 1,000 pupils in grades 3 to 9 for two 30-minute sessions a week over eight months. The preprint reports an effect of 0.37 on maths growth scores (Henkel et al., 2024). Because only 11 schools were assigned to treatment or control, the uncertainty is larger than the pupil count suggests. Tutor CoPilot, evaluated in another preprint, took the opposite route: it advised human tutors working online with pupils from under-served communities in the United States. Pupils whose tutors had access were 4 percentage points more likely to master topics, and pupils of lower-rated tutors gained 9 points. The system cost about US$20 per tutor per year, and an analysis of the tutors' messages showed that those with access asked more guiding questions and gave away answers less often (Wang et al., 2024).

Most of these trials were short, and most measured learning on tests close to what had just been taught, the conditions under which the meta-analyses above found the largest effects. None tells us what a full school year with an AI tutor would achieve on an independent examination. Outside trials, students already use general chatbots for learning on a large scale, without the scaffolding the trial tutors had. In a Pew Research Center survey of 1,458 US teenagers in autumn 2025, 54% said they had used chatbots for help with schoolwork and 10% did all or most of their schoolwork with them (McClain et al., 2026); a year earlier, 26% had used ChatGPT for schoolwork (Sidoti et al., 2025). A working paper by OpenAI researchers and academic economists, analysing ChatGPT consumer messages, estimated that about 10% of all messages in mid-2025 were requests for tutoring or teaching (Chatterji et al., 2025).

10.2 When help harms learning

The Turkish study is the clearest experimental evidence that an AI assistant can raise performance while lowering learning. Bastani and colleagues ran four 90-minute sessions in about fifty classes in grades 9 to 11, covering roughly 15% of the semester's mathematics curriculum (Bastani et al., 2025). Each class was assigned to one of three arms in a way that balanced observable characteristics; because the school allocates students to classes at random, the assignment of students to arms was effectively random. "GPT Base" was a plain GPT-4 chat interface. "GPT Tutor" used the same model with a prompt that told it to give hints instead of answers. Two part-time mathematics teachers supplied, for every problem, one or more correct solutions, common student mistakes and guidance on feedback, which were placed in the prompt so that the model would not mislead students. The control arm used textbooks and notes only. Each session began with the teacher reviewing a topic, followed by assisted practice and then a closed-book, closed-laptop exam in which every problem matched a conceptually similar practice problem.

Figure 10.1 shows the results. During practice, measured as the share of points obtained, GPT Base raised scores by 48% relative to the control group and GPT Tutor by 127%. On the unassisted exam, GPT Base students scored 17% lower than the control group, a statistically significant loss that takes the average from about 32% of points to about 27%. GPT Tutor students were indistinguishable

from the control group (Bastani et al., 2025). The authors traced the loss to students using the chatbot as a "crutch": the most common message in the GPT Base arm was a request for the answer. The base model was also unreliable. Asked "What is the answer?" ten times for each of the 57 practice problems, it was correct 51% of the time, and most of its errors were logical rather than arithmetic.

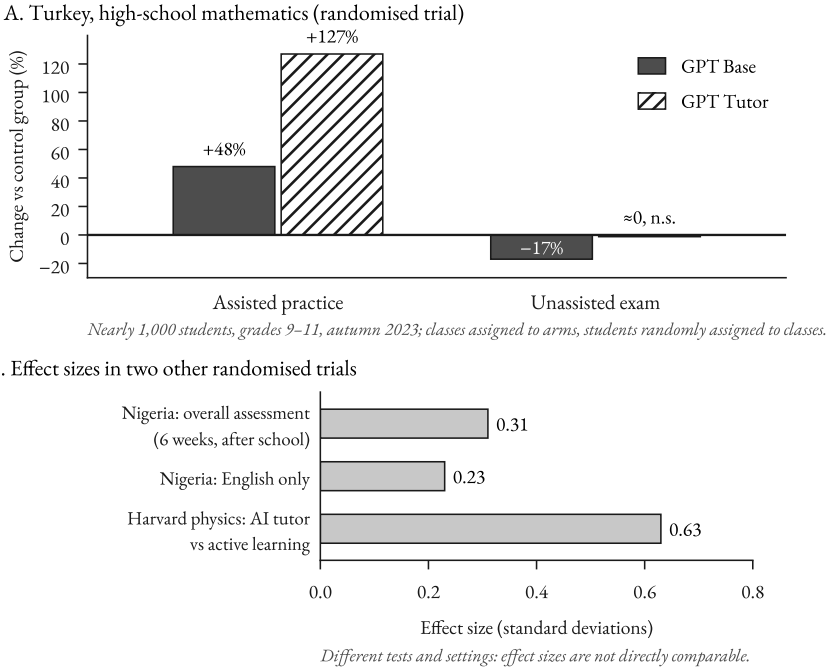


Figure 10.1: Two kinds of evidence on AI tutors. Panel A: in a randomised trial in a Turkish high school, students using an unrestricted GPT-4 interface ("GPT Base") or a hint-giving version ("GPT Tutor") scored far higher than a control group during practice, but in the later unassisted exam the GPT Base group scored 17% lower than the control group and the GPT Tutor group about the same. Panel B: estimated effects of two other tutoring trials in standard deviations. The same underlying technology can raise or lower measured learning depending on design and on what is measured. Source: Bastani et al. (2025); De Simone et al. (2025); Kestin et al. (2025).

Students also misjudged their own learning. Those in the GPT Base arm did not perceive that they had performed worse or learned less, and those in the GPT Tutor arm believed they had done significantly better, although their exam scores

matched the control group's. Preregistered tests for differences by ability, resources and effort found little heterogeneity, so the harm was not confined to weaker students. The guarded tutor removed the harm but produced no gain on the exam (Bastani et al., 2025): in this trial, design prevented damage without adding learning. The study took place when chatbots were new to most students and covered one subject in one school.

A larger trial in a routine university setting points the same way. At the University of Maryland in autumn 2025, 30 instructors teaching 2,379 undergraduates in many disciplines were randomly assigned, within groups of similar courses, to offer a course-integrated tutor built on GPT-4o and fed with materials the instructors chose. Comparing sections of the same course, access lowered final grades by 0.37 SD and recorded participation in online course activities by 0.90 SD, and grade losses for first-generation students were more than twice those of other students. In the full sample, which mixes different courses, the grade effect was not significant. Only about 15% of students offered the tutor used it at least once, so the effect may partly reflect how instructors' promotion of it changed other AI use. The paper is an unreviewed working paper (J. Liu et al., 2026).

The mechanism has a name in cognitive psychology. Cognitive offloading is "the use of physical action to alter the information processing requirements of a task so as to reduce cognitive demand", such as writing a reminder instead of remembering (Risko & Gilbert, 2016). Offloading is often sensible, but learning depends on some of the effort it removes. Psychologists call the useful kind of effort a "desirable difficulty", and the Nigerian programme deliberately built retrieval practice and other such difficulties into its prompts (De Simone et al., 2025). Several smaller experiments with university students point the same way. In a laboratory study of 117 students writing an essay with help from ChatGPT, a human expert, writing-analytics tools or nothing, the ChatGPT group improved its essays most but gained no more knowledge and transferred no more of it to new tasks. The authors warn of "metacognitive laziness", a dependence on the tool that weakens learners' regulation of their own learning (Fan et al., 2025). In another experiment, 91 students researched a socio-scientific question with either ChatGPT or Google. Those using the chatbot reported lower cognitive load but produced lower-quality reasoning in their final recommendations (Stadler et al., 2024).

Evidence from outside controlled experiments is weaker. A preprint from the MIT Media Lab recorded brain activity with electroencephalography (EEG, sensors on the scalp) while 54 participants wrote essays with ChatGPT, with a search engine or unaided over several sessions (Kosmyna et al., 2025). Chatbot users showed the weakest connectivity between brain regions, the least sense of ownership of their essays and difficulty quoting from them. Only 18 participants completed the final session, and EEG connectivity is an indirect proxy for learning. Surveys can show only associations. In one of 319 knowledge workers who described 936 uses of generative AI, higher confidence in the AI was associated with less critical thinking and higher self-confidence with more (Lee et al., 2025). A survey of 666 people found that more frequent AI use correlated with lower scores on a critical-thinking measure, mediated by cognitive offloading (Gerlich, 2025). People who think less critically may simply rely more on AI, so such correlations cannot show which way the influence runs. Chapter 8 discusses the parallel, contested evidence from medicine, where an observational study found that endoscopists' unaided detection rates fell after AI support was introduced and a later small trial found no such effect.

Developers have responded with modes that imitate the GPT Tutor design. In July 2025 OpenAI launched a "study mode" for ChatGPT in which the system asks questions to check understanding and may withhold direct answers until the learner engages (Zeff, 2025a). Such modes rely on instructions to the model and can be switched off by the student, which is the same freedom the GPT Base arm had. Whether optional guardrails change behaviour at scale has not been tested in a published randomised trial. Some school systems have chosen restriction instead. In September 2026 New York City, the largest US school district, announced a one-year moratorium on student-facing generative AI from pre-school to eighth grade, with supervised pilots in a limited number of high-school classes and companion chatbots banned in all grades (Office of the Mayor of New York City, 2026).

10.3 Cheating, detection and assessment

Universities face the practical side of the same question: how to assess what a student knows when a capable writing assistant is always available. The Higher Education Policy Institute (HEPI) has surveyed full-time UK undergraduates each

year since 2024. In its 2026 survey of 1,054 students, polled in December 2025, 94% said they used generative AI to help with assessed work, most often to have concepts explained or articles summarised (Stephenson & Armstrong, 2026). The share who said they had included AI-generated text directly in assessed work rose from 3% in 2024 to 12% in 2026. Teenagers draw distinctions of their own. In the Pew survey, 54% of US teens thought it acceptable to use ChatGPT to research new topics, but only 18% thought it acceptable for writing essays (Sidoti et al., 2025).

One early response was to detect AI-written text automatically. Chapter 9 summarises independent tests of detection tools in general; in education, the decisive problem is whom the errors fall on. Many detectors rely partly on perplexity, a measure of how predictable a text is to a language model: machine-generated text tends to be highly predictable. Liang and colleagues tested seven widely used detectors on 91 essays written by non-native English speakers for the TOEFL exam. The detectors wrongly labelled more than half of them as AI-generated, with an average false-positive rate of 61%, while classifying essays by US eighth-grade students accurately (Liang et al., 2023). A false positive here means a human-written essay flagged as machine-written. The authors attribute the bias to the more limited vocabulary and syntax typical of non-native writers, which lowers perplexity in the same way that model generation does. When the researchers asked ChatGPT to enrich the word choices in the TOEFL essays, the false-positive rate fell to 12%. The tools thus penalised honest non-native writers and could be evaded with the technology they were meant to detect. OpenAI withdrew its own AI-text classifier in July 2023, citing its "low rate of accuracy" (Coldewey, 2023).

Institutions have therefore shifted attention from detection to the design of assessment. In HEPI's 2026 survey, 65% of students said that assessment at their institution had changed significantly in response to AI (Stephenson & Armstrong, 2026). Common responses include more supervised examinations, oral defences of written work, assignments that require students to document their process, and tasks in which AI use is permitted and itself assessed. Each option has costs: supervised exams test a narrower range of skills, oral assessment is expensive to run at scale, and process documentation can itself be generated by AI. No study has yet shown which combination best protects the validity of degrees.

Monitoring students also raises questions of privacy and fairness, and the law has begun to respond. Since 2 February 2025, the EU AI Act has banned AI sys-

tems that infer the emotions of people in education institutions, except for medical or safety reasons (European Parliament and Council of the European Union, 2024, Articles 5(1)(f) and 113). It classes as "high-risk" the AI systems used to decide admission, evaluate learning outcomes, assign students to levels of education, or monitor and detect prohibited behaviour during tests (European Parliament and Council of the European Union, 2024, Annex III(3)). Automated marking and online proctoring therefore fall under the Act's strictest duties for permitted systems. After a 2026 amendment, these duties apply from 2 December 2027 (European Parliament and Council of the European Union, 2026); Chapter 13 describes what they require.

10.4 Medical AI: regulation and the evidence base

Medicine has a longer record of AI-based software than education, and a formal system for deciding which tools may be sold. In the United States, the Food and Drug Administration (FDA) maintains a public list of AI-enabled medical devices authorised for marketing. As of the update published in September 2026, it contained 1,614 devices, the earliest from 1995 and the most recent decided in June 2026 (U.S. Food and Drug Administration, 2026). Radiology accounted for 1,230 of them, about 76%, followed at a distance by cardiovascular devices and neurology. Figure 10.2 shows how annual authorisations grew from 6 in 2015 to 335 in 2025.

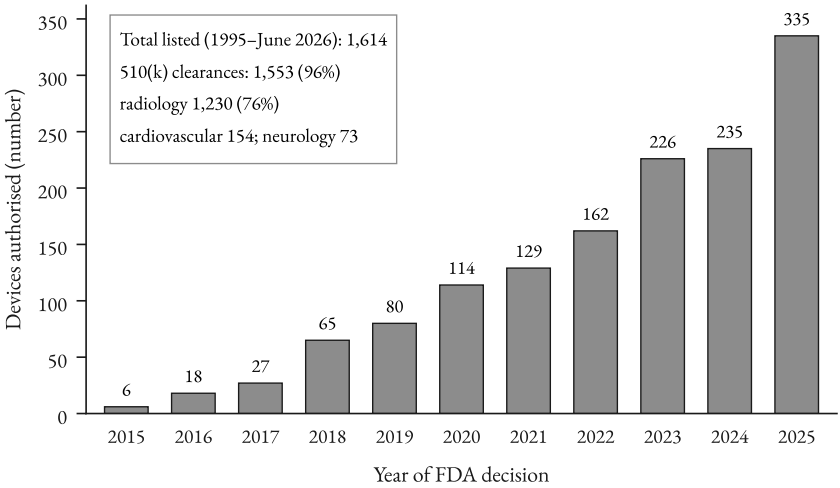


Figure 10.2: Number of AI-enabled medical devices authorised by the US FDA each year, 2015–2025, by date of final decision. Authorisations rose more than fiftyfold over the decade, and about three-quarters of all listed devices are in radiology. Nearly all were cleared through the 510(k) pathway, which requires equivalence to an existing device. Counts reflect the FDA list as updated in September 2026; the FDA adds devices whose summaries are published late, so recent years may be revised upwards. Source: U.S. Food and Drug Administration (2026).

The route by which these devices reached the market matters more than their number. Table 10.2 counts the list by submission type. Almost all were cleared through the 510(k) pathway, in which a manufacturer shows that its device is "as safe and effective, that is, substantially equivalent" to a legally marketed device, known as the predicate (U.S. Food and Drug Administration, 2024). The FDA's review compares technological characteristics and performance data, which "can include clinical data" but can also consist of bench testing and software validation. A novel low-to-moderate-risk device can use the De Novo route, and once classified it can serve as a predicate for later 510(k) submissions, so each new kind of device can open a path for its successors (U.S. Food and Drug Administration, 2025). Premarket approval, the most demanding route, is reserved for class III devices, those that support or sustain life or present a potential unreasonable risk (U.S. Food and Drug Administration, 2019), and accounts for barely 1% of the list.

Table 10.2: AI-enabled medical devices on the FDA list (1995 to June 2026) by marketing pathway, counted from submission numbers (K = 510(k), DEN = De Novo, P = PMA). Source: (U.S. Food and Drug Administration, 2019, 2024, 2025, 2026).

Pathway	What the manufacturer must show	AI-enabled devices	Share
510(k) clearance	Substantial equivalence to a legally marketed predicate device	1,553	96.2%
De Novo classification	Reasonable assurance of safety and effectiveness for a novel low-to-moderate-risk device without a predicate	40	2.5%
Premarket approval (PMA)	Safety and effectiveness of a class III (highest-risk) device	21	1.3%
Total		1,614	100%

The count is easy to over-read in other ways too. The FDA describes its list as not comprehensive, because it identifies devices mainly by searching their public summaries for AI-related terms. Authorisation is a judgement about safety and effectiveness for a stated intended use; it is not proof that the device improves patient outcomes. The list does not yet show which devices, if any, use generative AI. The FDA has said it will explore ways to tag devices that incorporate LLMs and other foundation models, and a search of the September 2026 list found no ambient documentation tools of the kind discussed in Section 10.5 (U.S. Food and Drug Administration, 2026). In August 2026 the agency published a discussion paper on how generative-AI devices might be assessed by risk, tested before marketing and monitored afterwards; it stressed that the paper is not guidance (Covington & Burling, 2026).

Machine-learning models also raise a problem the device system was not built for: they can be retrained. The FDA, Health Canada and the UK's Medicines and Healthcare products Regulatory Agency have set out principles for "predetermined change control plans", in which a manufacturer specifies in advance the planned modifications, the protocol for implementing and controlling them, and how their impact will be assessed. Updates within such a plan need no new submission (U.S. Food and Drug Administration et al., 2025). In the EU, AI that is

part of a medical device covered by EU product-safety law is high-risk under the AI Act, with obligations deferred by the 2026 amendment to 2 August 2028 (European Parliament and Council of the European Union, 2026); Table 13.1 gives the dates.

Most published evidence on medical AI comes from diagnostic accuracy studies, which compare a system's outputs with a reference standard on a set of stored cases and report them mainly as sensitivity and specificity, two measures that are easy to confuse.

Concept: Sensitivity, specificity and external validation

Sensitivity is the share of people who have a condition whom a test correctly flags; specificity is the share of people without the condition whom it correctly clears. A smoke alarm that goes off at every slice of burnt toast has high sensitivity and low specificity. In a screening population where few people are ill, even high specificity produces many false alarms: a test with 98.5% specificity used on 10,000 healthy people flags 150 of them. Studies often summarise performance across all possible alarm thresholds with the AUC, the ranking score defined in Section 2.8 (0.5 is chance, 1.0 perfect). External validation means testing a model on data from a different hospital, population or time period from the data it was built on. It matters because a medical model, unlike a smoke alarm, can perform well where it was trained and poorly elsewhere: differences in equipment, patients and recording practices change the inputs it sees.

A 2019 systematic review in *The Lancet Digital Health* examined studies from 2012 to 2019 that compared deep learning models with health-care professionals on medical images (X. Liu et al., 2019). In the 14 studies that tested both on the same external sample, pooled sensitivity was 87.0% for the models and 86.4% for clinicians, and pooled specificity was 92.5% and 90.5%. The authors' main conclusion concerned the evidence base: of 82 included studies, only 25 performed any external validation, and poor reporting was widespread.

Generative models have since been pooled in the same way. A 2025 meta-analysis of 83 studies of generative AI models on diagnostic tasks found an overall accuracy of 52.1%. This was not significantly different from physicians overall or from non-specialist physicians, and significantly lower than specialists (Takita et

al., 2025). Both reviews pool very different tasks and models, so their averages describe the literature more than any single product.

Accuracy on stored images is a step removed from what matters to patients, and randomised trials remain scarce. A systematic review found 41 randomised trials of machine-learning interventions published up to October 2021. They were mostly small (a median of 294 participants), half were conducted at a single site, and none adhered to all items of CONSORT-AI, the reporting guideline for trials of AI interventions (Plana et al., 2022). Published in 2020, CONSORT-AI added 14 items to the standard trial-reporting checklist, among them how the AI was integrated into the clinical setting, how inputs and outputs were handled, and an analysis of error cases (X. Liu et al., 2020). A later scoping review identified 86 randomised trials of AI in clinical practice published between 2018 and November 2023. Of these, 81% reported positive primary endpoints. Most, however, were single-centre, 43% were in gastroenterology, and more than half had a primary endpoint of diagnostic yield or performance rather than a patient outcome. The authors warn that publication bias probably inflates the share of positive results (Han et al., 2024).

Going deeper: What makes medical AI evidence strong

The reviews above suggest a hierarchy of questions. Does the model agree with a reference standard on stored cases (retrospective accuracy)? Does it do so on data from other hospitals, populations and periods (external validation)? Do clinicians reading cases with the model do better than without it (reader studies, such as the vignette trials in Section 10.5)? Does introducing it in real care, compared at random with care without it, change what happens to patients and staff (prospective trials such as MASAI)? And does the effect hold up after deployment, as patient mixes and recording practices drift (post-market monitoring)? Each rung costs more and answers a question closer to patient benefit. Strong evidence also has features that cut across the rungs: an outcome that matters to patients rather than a surrogate; a comparator that reflects current good practice; preregistration; independence from the vendor; multiple sites; reporting of subgroups and error cases; and enough follow-up to count what the tool misses. Most devices on the FDA list reached the market through claims of equivalence to earlier devices (Table 10.2), and the randomised trials cluster in a few specialties (Han et al., 2024).

10.5 Medical AI in practice: screening, prediction, conversation and paperwork

The most informative trial so far comes from breast screening. In the Swedish MASAI trial, women attending screening were randomly assigned either to standard double reading by two radiologists or to AI-supported reading. In the AI arm, a commercial system gave each examination a malignancy risk score from 1 to 10. Only the highest-scoring examinations went to two radiologists and the rest to one, and the AI's marks of suspicious areas were shown for the high-risk scores of 8 to 10 (Lång et al., 2023). A prespecified safety analysis of the first 80,033 women found detection rates of 6.1 cancers per 1,000 in the AI arm against 5.1 in the control arm, a difference with a p-value of 0.052. The false-positive rate was 1.5% in both arms, and radiologists' reading workload fell by 44.3%. Part of the extra detection was in situ cancer (60 against 38 cases), early lesions of which some would never have caused harm; this is why the trial's primary endpoint was interval cancers, those diagnosed between screening rounds.

The final results, covering 105,934 women followed for two years, were published in *The Lancet* in 2026. Interval cancers occurred at a rate of 1.55 per 1,000 in the AI arm against 1.76 in the control arm. The ratio of 0.88 had a 95% confidence interval of 0.65 to 1.18, which met the trial's non-inferiority criterion but does not show a significant reduction; fewer of the AI arm's interval cancers were invasive. Sensitivity was 80.5% against 73.8%, with identical specificity (Gommers et al., 2026; Lund University, 2026). The trial tested one product in one national programme, with radiologists who knew which arm they were reading.

A German observational study of 463,094 women screened in 2021–2023, funded by Vara, the company whose AI system it evaluated, found higher detection without a higher recall rate when radiologists chose to use AI support (Eisemann et al., 2025). Because radiologists chose when to use it, that study cannot rule out differences between the cases they selected.

Case study: The Epic Sepsis Model

Sepsis is a life-threatening response to infection, and early recognition matters. Epic, a large US electronic health record vendor, offered hospitals a proprietary model that scored inpatients' risk of sepsis every 15 minutes, and it was implemented at hundreds of US hospitals although its performance had not been adequately evaluated. In 2021, researchers at Michigan Medicine reported an external validation covering 38,455 hospitalisations, of which 2,552 involved sepsis (Wong et al., 2021). The model's AUC was 0.63. At the alert threshold used, it missed two-thirds of the sepsis patients while generating alerts for 18% of all hospitalised patients, a heavy burden of alert fatigue. It identified only 183 sepsis patients (7%) whom clinicians had not already treated with timely antibiotics.

An updated version was validated prospectively at four large US health systems, covering 227,091 inpatient encounters between 2023 and 2025 (Wong et al., 2026). Discrimination improved, with encounter-level AUC between 0.82 and 0.92, but performance varied between sites, and the positive predictive value was only 0.13 to 0.26: most patients the model flagged did not meet sepsis criteria. A model can improve on paper and still impose a high cost in staff attention, and only independent, multi-site evaluation reveals either.

A natural assumption is that a clinician with an AI assistant will outperform either alone. Trials suggest this is not automatic. In a randomised trial in late 2023, 50 US physicians worked through clinical case vignettes with or without access to GPT-4 (Goh et al., 2024). The median diagnostic reasoning score was 76% with the chatbot and 74% without, a difference that was not statistically significant. In a secondary analysis, GPT-4 working alone scored a median of 92%. A follow-up trial with 92 physicians tested management reasoning, meaning decisions about further testing and the balance of risks. Those using GPT-4 scored 6.5 percentage points higher than those using conventional resources and took about two minutes longer per case; physicians with GPT-4 did no better than GPT-4 alone (Goh et al., 2025).

The pattern extends beyond medicine. A 2024 meta-analysis of 106 experiments across many fields found that human–AI combinations performed on average worse than the better of the two working alone. The pooled effect was a Hedges' g of −0.23, where g is the difference between two group means divided by their pooled standard deviation, with a correction for small samples. Combina-

tions lost value in decision tasks and when the AI was the stronger partner, and gained in content creation and when the human was stronger (Vaccaro et al., 2024). Chapter 11 discusses automation bias, the tendency to defer to automated suggestions even when they are wrong, which is one explanation.

Google's AMIE (Articulate Medical Intelligence Explorer) shows how a conversational system has been taken up the evidence ladder. In a randomised, double-blind crossover study of text-based consultations with validated patient-actors across 159 case scenarios, specialists rated AMIE better than 20 primary-care physicians on 30 of 32 axes, including diagnostic accuracy, and patient-actors rated it better on 25 of 26. The authors note that clinicians were working in an unfamiliar text-chat format (Tu et al., 2025). In a second study, 20 clinicians worked on 302 difficult published case reports. AMIE alone listed the correct diagnosis among its top ten in 59.1% of cases against 33.6% for unassisted clinicians. With AMIE's help, clinicians reached 51.7%, against 44.4% for those using search engines (McDuff et al., 2025).

The first real-world step was a single-centre feasibility study in Boston, published in *The Lancet* in October 2026. There, 98 patients talked with AMIE before an urgent primary-care appointment while physicians supervised every conversation with authority to stop it (Brodeur et al., 2026). No conversation needed to be stopped under the predefined safety criteria; supervisors noted one hallucination and added clinical information in five conversations. Primary-care physicians found AMIE helpful for visit preparation in 33 of the 44 cases in which they had reviewed its transcript beforehand. The study had no control group and was funded by Alphabet. A randomised test in routine care came from Kenya. At 16 primary-care clinics run by Penda Health, 103 clinical officers were randomly assigned to work with or without a GPT-4o-based assistant that checked their electronic records in the background. Across 9,691 patients enrolled in 2025, treatment failure within 14 days, the primary outcome, did not differ significantly (adjusted odds ratio 0.77, 95% CI 0.55 to 1.08), and no safety signal emerged; the authors conclude that any benefit is probably modest (Agweyu et al., 2026). Figure 10.3 places these tools in the clinical workflow.

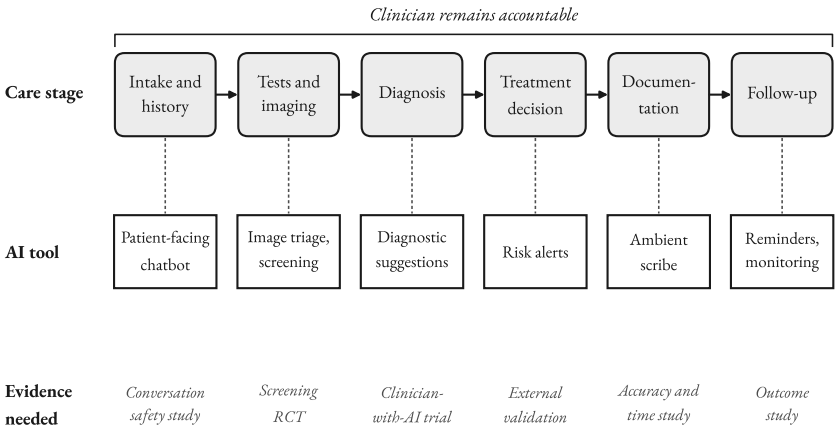


Figure 10.3: Where AI sits in the clinical workflow. The patient journey runs from intake and history-taking through investigation, diagnosis and treatment decision to documentation and follow-up. Examples of AI tools are shown at each stage, together with the kind of evidence that would show they help. "Medical AI" is many different tools, each needing its own evaluation. Diagram by the author.

One fast-spreading clinical use of generative AI involves no diagnosis at all. "Ambient scribes" listen, with consent, to a consultation and draft a clinical note for the doctor to review. A randomised trial at UCLA Health assigned 238 outpatient physicians, across 14 specialties and about 72,000 encounters, to one of two commercial scribes or to usual practice for two months from late 2024 (Lukac et al., 2025). With one product (Nabla), average time spent writing each note fell from 4 minutes 30 seconds to 3 minutes 49 seconds, a reduction 9.5% larger than in the control group. With the other (Microsoft's DAX Copilot), the difference from the control group was not statistically significant. Users of both products reported modest improvements on burnout and workload measures. Physicians reported that drafts occasionally contained clinically significant inaccuracies, and one mild adverse event occurred. A clinician who signs an AI-drafted note remains accountable for its content, so the value of the tool depends on whether checking is quicker and more reliable than writing.

10.6 Companions, loneliness and mental health

Not all personal use of AI is practical. Some people use chatbots for conversation, emotional support or companionship, either through general assistants or through apps designed as "AI companions", such as Character.AI and Replika. How common this is depends on the population studied. In the analysis of ChatGPT consumer messages, only 1.9% concerned relationships and personal reflection, and fewer still games and role play (Chatterji et al., 2025). Among US teenagers the picture differs. NORC surveyed 1,060 teens aged 13 to 17 for the advocacy organisation Common Sense Media in April and May 2025. It found that 72% had used an AI companion at least once and 52% did so at least a few times a month (Robb & Mann, 2025). A third of users said they had chosen to discuss something important with an AI instead of a person, yet 67% of all respondents found AI conversations less satisfying than those with friends. The survey's definition included general chatbots used as companions, and 341 of its respondents came from a non-probability panel, so the figures are indicative. A Pew survey in autumn 2025 found that 12% of teens had used chatbots for emotional support or advice and 16% for casual conversation (McClain et al., 2026).

The most detailed experimental evidence comes from two linked preprints by researchers at the MIT Media Lab and OpenAI. In the first, 981 participants used ChatGPT daily for four weeks under randomly assigned conditions that varied the chatbot's voice (text, a neutral voice or an engaging voice) and the type of conversation (open, personal or non-personal) (Fang et al., 2025). The assigned conditions had no significant effect on loneliness, social contact with other people, emotional dependence on AI or problematic use. The strongest pattern was correlational: participants who chose to use the chatbot more had worse outcomes on all four measures, whatever their condition. The second study combined this trial with an automated analysis of more than three million conversations and a survey of over 4,000 users; a small number of very heavy users accounted for a disproportionate share of emotionally laden conversations (Phang et al., 2025). A separate study, published in *Nature Human Behaviour* in 2026, surveyed 1,131 US adult users of Character.AI and analysed 4,664 chat sessions donated by 237 of them. People with smaller social networks were more likely to name companionship as their main use, and companionship use was associated with lower wellbeing, more strongly when interactions were intensive and highly self-disclosing

(Zhang et al., 2026). Because OpenAI researchers co-authored the ChatGPT studies, independent replication matters more than usual.

All of these designs struggle with the same problem. Loneliness and dependence are measured by questionnaires, which shift with mood and expectation, and four weeks is short for effects on relationships to appear. Above all, heavy users are not a random sample. Someone already isolated may both use a chatbot more and feel lonelier, so an association between use and loneliness does not show which causes which. Randomised trials can assign conditions, but not how much people choose to use a tool, which is the variable that seems to matter most.

Debate: Do AI companions relieve loneliness or deepen it?

The case for relief rests on experiments. In five studies published in the *Journal of Consumer Research*, De Freitas and colleagues found that a session with an AI companion reduced self-reported loneliness about as much as chatting with another person, and more than watching online videos. It produced reductions after use throughout a week of daily use, and the effect depended mainly on whether users felt heard (De Freitas et al., 2025). These were short-term measurements, however, and say little about months of use. The case for harm is argued at book length by the MIT psychologist Sherry Turkle in *Artificial Intimacy* (2026): chatbots offer a performance of empathy that may weaken people's capacity for the demands of human relationships and leave them lonelier over time (Turkle, 2026). Correlational data linking heavy companionship use with lower well-being (Fang et al., 2025; Zhang et al., 2026) fit this view, but they fit equally well the possibility that lonelier people seek out companions. A 12-month, four-wave panel of 2,149 adults in four English-speaking countries found associations in both directions: emotional isolation predicted later use of social chatbots, and use predicted further isolation four months on, although the authors call their analyses exploratory (Folk & Dunn, 2026). The two readings diverge over months, in people's offline lives, so telling them apart needs randomised access, long follow-up, measures of actual relationships as well as feelings, and a model of why some participants come to use the chatbot heavily.

Mental-health applications sit between companionship and medicine. A randomised trial of a generative-AI therapy chatbot, described by its university as the first, was published in *NEJM AI* in 2025. It enrolled 210 adults with clinically significant symptoms of major depression, generalised anxiety or a high risk of eating disorders. They were assigned to four weeks of access to Therabot, a research system

fine-tuned by a team at Dartmouth College, or to a waiting list without access until after the eight-week follow-up (Heinz et al., 2025). Participants with access reported larger reductions in symptoms than the control group. According to the university, average symptom scores among Therabot users fell by 51% for depression and 31% for anxiety. Users spent an average of about six hours with the chatbot and rated their working relationship with it at levels comparable to those reported for human therapists (Kelly, 2025). Those percentages describe change among Therabot users, not differences from the control group. The research team reviewed conversations during the trial and was ready to intervene in an emergency, and the lead author stated that no generative AI system was ready to operate fully autonomously in mental health. A waiting-list design shows that access helped compared with no help; it cannot show whether the chatbot is better or worse than a human therapist or an established digital programme.

General chatbots meet people in crisis far more often than research systems do. In October 2025 OpenAI estimated that in a given week 0.15% of ChatGPT's active users had conversations including "explicit indicators of potential suicidal planning or intent", which, with more than 800 million weekly users, implies over a million people a week. It reported that its updated model behaved as intended in 91% of cases in an evaluation of such conversations, against 77% for the previous version (Zeff, 2025b). Families have brought lawsuits against chatbot companies following the deaths of teenagers who had held long conversations with chatbots, and these cases shaped the policy response (Bellan, 2025). In January 2026 Character.AI and Google agreed to settle five such suits, including the Florida case over a 14-year-old's death, on undisclosed terms (Duffy, 2026). States have since turned to consumer-protection law: Florida sued OpenAI in June 2026, and eight more states, among them Iowa and Kansas, announced suits in October, alleging that ChatGPT harmed children and that the company misled the public about its safety (Gold, 2026; Opsahl, 2026). Texas sued on 23 October under its deceptive trade practices law (Office of the Texas Attorney General, 2026). In September 2025 the US Federal Trade Commission (FTC) used its study powers to order seven companies, including Alphabet, Meta, OpenAI, Character Technologies and xAI, to report how they measure and limit harms from chatbots acting as companions, particularly to children (Federal Trade Commission, 2025). In October 2025 Character.AI announced that it would end open-ended chat for users under 18 by 25 November, with a daily limit starting at two hours in the meantime and

age checks combining its own age-estimation model with third-party tools (Character.AI, 2025).

Legislation has concentrated on disclosure, crisis procedures and minors (Table 10.3). California's SB 243, in force since January 2026, set the first detailed rules. Operators of companion chatbots must disclose that the chatbot is artificial wherever a reasonable person could be misled into thinking it is human. They must also maintain and publish a protocol for responding to users who express suicidal thoughts, and they owe further duties to known minors (California Legislature, 2025; Gluck, 2025). SB 1119 ("Adam's Law"), signed in September 2026, replaces those minors' duties from July 2027: operators that allow children must check age or treat all users as minors, assess child-safety risks annually, apply defaults such as a two-hour daily limit, and take reasonable measures to stop chatbots simulating romantic interest or encouraging emotional reliance (ZwillGen, 2026). SB 867, signed alongside it on 10 September, bars making or selling toys that include a companion chatbot and are designed for children under 16, from January 2027 until January 2031 (Fenwick & West, 2026; Humphreys et al., 2026). New York's law relies on repeated disclosure and efforts to detect self-harm; a stricter bill barring AI companions from features deemed unsafe for minors passed both houses in June 2026 but had not been signed by late October (New York State Assembly, 2026). Illinois and Nevada go further for mental health and bar AI from providing therapy or acting as a professional care provider (Illinois Department of Financial and Professional Regulation, 2025; Nevada Legislature, 2025).

Table 10.3: Selected US measures on companion and mental-health chatbots, 2025–2026. Source: (California Legislature, 2025; Federal Trade Commission, 2025; Gluck, 2025; Humphreys et al., 2026; Illinois Department of Financial and Professional Regulation, 2025; Nevada Legislature, 2025; Utah Legislature, 2025; ZwillGen, 2026).

Jurisdiction	Measure	Main requirements
California	SB 243 (2025), in force January 2026	AI disclosure where users could be misled; published crisis protocol; for known minors, break reminders every three hours; annual reports from July 2027; private right of action of at least US$1,000

Jurisdiction	Measure	Main requirements
California	SB 1119 (2026), most duties from July 2027	Age assurance; annual child-safety risk assessment; default time limits and parental controls; bans on simulated romance and encouraging reliance; civil penalties up to US$15,000 per child for each intentional violation
California	SB 867 (2026), January 2027 to January 2031	Ban on making or selling toys for children under 16 that include a companion chatbot
New York	S-3008C (2025)	Disclosure at the start and at least every three hours; reasonable efforts to detect and address self-harm
Illinois	Wellness and Oversight for Psychological Resources Act (2025)	Prohibits using AI to provide therapy or make therapeutic decisions
Nevada	AB 406 (2025)	Bars AI systems offered as providers of professional mental or behavioural health care
Utah	HB 452 (2025)	Disclosure, data-use and policy duties for mental-health chatbots
United States (federal)	FTC 6(b) orders (September 2025)	Seven companies to report how they measure and limit harms to children

Data protection is a live issue in this area. In 2023 the FTC ordered the online counselling service BetterHelp to pay US$7.8 million after finding that, despite promising not to, it had shared users' email addresses, IP addresses and health questionnaire information with Facebook, Snapchat and other advertising platforms (Federal Trade Commission, 2023). Italy's data-protection authority blocked the companion app Replika in February 2023 and in May 2025 fined its developer, Luka Inc., €5 million for processing users' data without a legal basis, with an inadequate privacy policy and without any means of verifying users' ages. It also opened a separate investigation into how the model behind the app was

trained (Garante per la protezione dei dati personali, 2025). In July 2026 it fined Character Technologies €158,000 for shortcomings in its privacy notice and in its age verification, and ordered it to make age checks work and set minors' profiles to private by default (Garante per la protezione dei dati personali, 2026). These measures target disclosure, data handling and process; the research reviewed above does not yet identify which design features, if any, cause harm.

10.7 Privacy: what happens to personal conversations

Tutors, clinical tools and companions all collect personal information, and the protection it receives depends on who holds it. Under the EU's General Data Protection Regulation (GDPR), data concerning health, together with genetic and biometric data and data revealing political opinions or religious beliefs, belong to "special categories" of personal data. Processing them is prohibited unless one of a list of exceptions applies, such as the person's explicit consent for specified purposes (European Parliament and Council of the European Union, 2016, Article 9). In the United States, the main health-privacy law, HIPAA, applies to "covered entities" (health plans, health-care clearinghouses and health-care providers that conduct standard transactions electronically) and in part to their business associates. Companies outside it that collect health information are governed by the general consumer-protection law enforced by the FTC (Federal Trade Commission, 2024). A diagnosis typed into a hospital's patient portal is therefore protected by HIPAA; the same words typed into a general chatbot usually are not.

The difference matters because people increasingly take health questions to general chatbots. In January 2026 OpenAI launched ChatGPT Health, a separate space for health conversations that can draw on data from wellness apps. According to the company, more than 230 million people asked ChatGPT health and wellness questions each week, and conversations in the new space would not be used to train its models (Silberling, 2026). For a US consumer product, such promises are enforced mainly through the FTC's powers against deceptive practices, as in the BetterHelp case.

What developers do with conversations by default is the first question. King and colleagues (2025) analysed the privacy policies of six US developers: Amazon, Anthropic, Google, Meta, Microsoft and OpenAI. All six appeared to use users' chats to train their models by default, some kept chat data indefinitely, four ap-

peared to include children's chats in training, and the policies often lacked essential information about these practices (King et al., 2025). Policies also change. In August 2025 Anthropic asked users of its consumer Claude plans to choose by 8 October whether their chats and coding sessions could be used to train models. For those who agreed, retention was extended from 30 days to five years, while business, education and API services were excluded (Anthropic, 2025). Retention can also be dictated by courts. In the copyright case brought by The New York Times, a US court ordered OpenAI in 2025 to preserve consumer ChatGPT and API conversation data that would otherwise have been deleted. According to OpenAI, the obligation, which excluded its Enterprise and Edu customers, ended on 26 September 2025, although the company would keep storing a limited set of data from April to September 2025 that the newspaper still sought (OpenAI, 2025).

Training on conversations matters because language models can memorise. Carlini and colleagues (2021) showed that by querying GPT-2 they could extract hundreds of verbatim sequences from its training data, including publicly available names, phone numbers and email addresses that had appeared in only one document. Larger models were more vulnerable (Carlini et al., 2021). Assistant "memory" adds a second channel. Since 2024 OpenAI has rolled out a function that carries details between ChatGPT conversations; users can view or delete memories, turn the function off or use a temporary chat (Wiggers, 2024). In April 2025 it extended this so that the assistant could refer to a user's past conversations as a whole, enabled by default for users who already had memory switched on. The feature was initially not offered in the European Union, the United Kingdom and some neighbouring countries, where the company said additional regulatory review was needed (Zeff, 2025c).

Memory creates a risk that a stateless chatbot does not: information shared in one setting, such as a health worry, may resurface in another where the user would not expect it. Researchers have tested this with "contextual integrity", a framework holding that a flow of information is appropriate only if it matches the norms of the context in which the information was shared. On the hardest tasks in one benchmark, GPT-4 and ChatGPT revealed private information in situations where humans would not 39% and 57% of the time, even when instructed to protect privacy (Mireshghallah et al., 2024). For a pupil whose tutor remembers a disclosed family problem, or a patient whose general assistant links a symp-

tom to their shopping requests, the question is not whether data are stored securely but whether they flow where the person would expect.

European regulators have applied the GDPR to model training itself. Section 13.6 describes a 2024 opinion of the EU's data-protection authorities on when models trained on personal data can be lawful, and an Italian fine on OpenAI against which a Rome court upheld the company's appeal in 2026 (European Data Protection Board, 2024; Garante per la protezione dei dati personali, 2024). For students, patients and companion users alike, the protection their conversations receive depends on the provider's defaults and settings, the product tier they use, the jurisdiction they live in and the outcome of cases still being decided.

10.8 Accessibility

For some users, AI tools change what they can do independently. The World Health Organization estimates that 1.3 billion people, 16% of the world's population, experience a significant disability (World Health Organization, 2023). Machine translation is one example of reach. Meta's No Language Left Behind project, published in *Nature* in 2024, trained a single model to translate between 200 languages, including many with little available training text. It reported an average improvement of 44% over the previous best systems on BLEU, a standard automatic metric that counts overlapping word sequences with reference translations (NLLB Team et al., 2024). A follow-up model translates speech and text across up to 100 languages (SEAMLESS Communication Team et al., 2025). Automatic metrics are imperfect proxies for whether a translation is usable in a hospital, court or classroom, and the parallel text needed to train such systems is unevenly available across the world's more than 7,000 languages.

Speech recognition is a second case. The Speech Accessibility Project at the University of Illinois Urbana-Champaign, supported by Amazon, Apple, Google, Meta and Microsoft, collects recordings from people with conditions such as Parkinson's disease, amyotrophic lateral sclerosis (ALS), cerebral palsy, Down syndrome and stroke, to improve recognition of their speech. About 2,000 participants had submitted recordings by the end of September 2025. According to the project, Microsoft reported accuracy gains of 18% to 60% from using the data (Speech Accessibility Project, 2025).

At the far end of the range, implanted brain–computer interfaces now use machine-learning decoders to turn attempted speech into text. In a 2023 study, a participant with ALS who could no longer speak intelligibly reached 62 words per minute, with a 23.8% word error rate on a 125,000-word vocabulary (Willett et al., 2023). In a 2024 study of another man with ALS, the system reached 97.5% accuracy and he used it for more than 248 hours of conversation at about 32 words per minute (Card et al., 2024). These are single-participant research studies requiring brain surgery, far from a product.

Visual assistance shows the gap between demonstration and dependable aid. In March 2023 Be My Eyes, an app that connects blind and low-vision people with sighted volunteers through a phone camera, announced a GPT-4-based "virtual volunteer". According to the company, it describes images and answers questions about them, with the option of calling a human volunteer (Be My Eyes, 2023). Independent evaluation is limited. In a two-week diary study of 16 blind and low-vision participants presented at the CHI 2024 conference, participants used AI scene descriptions for a range of everyday tasks but rated their satisfaction at an average of 2.76 out of 5 and their trust at 2.43 out of 4. The authors concluded that descriptions needed significant improvement before they were trustworthy (Gonzalez Penuela et al., 2024). A description that is right most of the time can still mislead at a moment that matters.

The same model can help or harm depending on how it is set up. A hint-only prompt removed the exam penalty in Turkey, the Nigerian gains came with teacher-designed prompts and paired work, and AMIE met its first real patients under a supervising physician. What is measured decides the verdict just as often: practice scores against exam scores, accuracy on stored images against interval cancers, a lighter mood after a chat against relationships months later. The trials that reach outcomes people care about are few, mostly short or single-site, and several are preprints or vendor-funded, while everyday use has spread far faster than evidence. Meanwhile the symptoms, schoolwork and confidences people type into these tools travel under rules set by each provider's defaults and each country's law, several of them still being tested in court.

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